

Elucidating Surface and Bulk Phase Transformation in Fischer–Tropsch Synthesis Catalysts and Their Influences on Catalytic Performance

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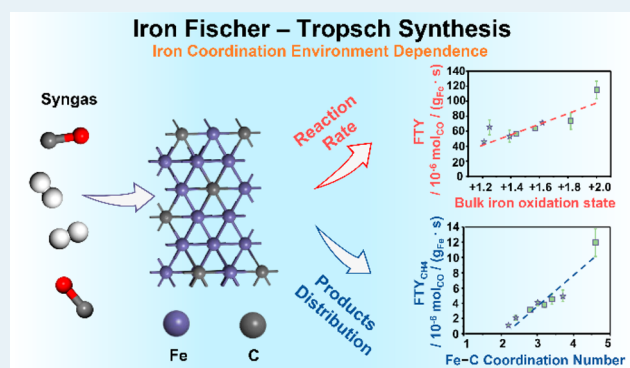
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Supporting Information

ABSTRACT: Understanding the structure–performance relationship for Fe-based Fischer–Tropsch synthesis (FTS) catalysts is crucial for the design of improved FTS catalysts. Herein, catalysts derived from four pure phases of iron (i.e., nanosized Fe_2O_3 , $\chi\text{-Fe}_5\text{C}_2$, $\theta\text{-Fe}_3\text{C}$, and Fe^0) were obtained under controlled conditions. Combining various surface and bulk characterization techniques, we revealed that core–shell structures were formed for all catalysts irrespective of the catalyst precursors, while the compositions within the core and shell strongly depend on the intrinsic properties of the precursors employed. Most importantly, we report that the iron coordination environment mainly governs the performance during FTS. The iron time yield is found to correlate linearly with the average bulk iron oxidation state rather than the surface properties, and the CH_4 yield is related to the coordination number of the Fe–C bond. This study thus provides general descriptors to interpret the structure–performance relationships of Fe-based FTS catalysts.

KEYWORDS: Fischer–Tropsch synthesis, iron catalysts, structure–performance relationship, descriptor, iron oxidation state, Fe–C coordination number



INTRODUCTION

The Fischer–Tropsch synthesis (FTS) is a process used to convert syngas directly to hydrocarbon fuels and chemical feedstocks,¹ which offers feedstock flexibility to meet a rapid increase of energy demand.² Currently utilized, iron-based catalysts exhibit significant advantages including their low price, chemical robustness, high tolerance for the wide H_2/CO ratios obtained for H_2 -deficient syngas obtained from coal, and good biomass conversion. Generally, iron oxides have been used as catalyst precursors for FTS, undergoing reduction and carburization during activation and reaction. But a complex mixture of phases (e.g., Fe oxides and carbides) is often experimentally identified under FTS conditions. Researchers including de Smit et al. claimed that the stability and interconversion of these carbides strongly depend on the carbon chemical potential and reaction temperature.³ Furthermore, the carbide phases could be reoxidized in a syngas atmosphere. The rich complexity of iron species and the dynamic nature of the iron–carbon–oxygen system under the FTS environment makes it challenging to identify and deconvolute the various factors affecting the structure–performance relationship in these systems.^{4–6} The air-

sensitivity of iron species also puts forward higher requirements for characterization techniques.⁴ Despite tremendous effort being made to elucidate the role of different iron carbides, the results presented in the literature are somewhat controversial. The Hägg carbide ($\chi\text{-Fe}_5\text{C}_2$) is mostly commonly regarded as the primary active species, as evidenced by the relationship between $\chi\text{-Fe}_5\text{C}_2$ and catalytic activity in a number of cases.^{3,6,7} Fe_2C is predicted to be more active in CO dissociation and hydrogenation by density functional theory (DFT) calculations, but it is rarely studied due to its thermodynamic instability.^{8–10} $\theta\text{-Fe}_3\text{C}$ has traditionally been considered as a deactivation phase;^{3,5} however, Liu et al. have found it contributes to the high selectivity of lower olefins.¹¹ More recently, Fe_7C_3 was found to have the highest turnover frequency compared to the $\varepsilon\text{-Fe}_2\text{C}$ and $\chi\text{-Fe}_5\text{C}_2$ phases.¹² Therefore, the speciation of iron carbides goes beyond the single-site paradigm, which includes chemical transformation and is influenced by precursor composition, synthesis method,

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Scheme 1. Illustration of Sample Collection under Two Conditions

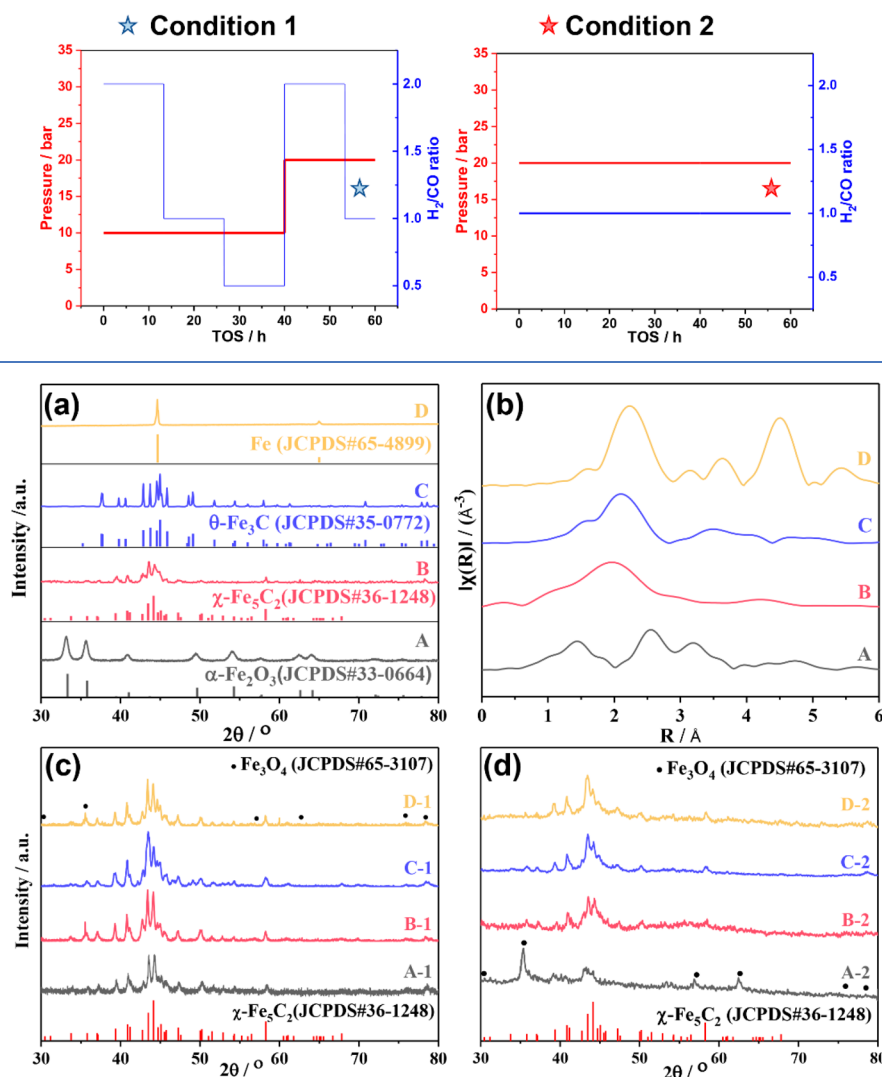


Figure 1. Characterization of catalysts precursors and used catalysts: (a,b) experimentally obtained XRD patterns and k^2 -weighted Fourier-transformed Fe K-edge EXAFS data of precursors; (c,d) XRD patterns of used catalysts under condition 1 and condition 2.

pretreatment, and reaction conditions. Besides the complexity and instability of iron phase, the structure-sensitivity of FTS reaction also plays an important role in catalytic performance. Size-dependence has been extensively reported by experimental work, which affects the coordination environment of Fe atoms,^{13,14} and different planes of a certain iron phase also show different CO activation and C–C coupling abilities in the DFT study.^{15–17} Third, the heterogeneous structure of Fe catalysts from surface to bulk is observed under FTS atmosphere.^{18–21} These issues make it challenging to interpret the structure–activity relationship generally. On the other hand, the products selectivity in FTS is of great scientific and industrial importance. Generally, the products distribution is a compromise between C–C coupling and secondary hydrogenation, which is closely related to the chemical properties of the Fe species. Insight into the relationship between the Fe species and selectivity is highly desirable for designing selective catalysts of FTS.

Herein, we initiated a study with four pure phases of iron (Fe_2O_3 , $\chi\text{-Fe}_5\text{C}_2$, $\theta\text{-Fe}_3\text{C}$, and Fe^0 ; denoted as A, B, C, and D, respectively) as precursors, and collected two series of derived

catalysts at 60 h under the FTS conditions ($T = 340^\circ\text{C}$, $\text{H}_2/\text{CO} = 1$, 20 bar, $\text{WHSV} = 30\,000\text{ mL}/(\text{g}_{\text{cat}}\cdot\text{h})$) with the Flowrence high-throughput unit.²² As shown in Scheme 1, series 1 catalysts (denoted as A-1, B-1, C-1, and D-1) were subject to lower pressures and varied H_2/CO ratios before reaching the FTS conditions; series 2 catalysts (denoted as A-2, B-2, C-2, and D-2) were pretreated by keeping the FTS conditions. By means of comprehensive and complementary “quasi in situ” characterization, we aimed at correlating the surface and bulk properties of the intermediate samples with both activity and selectivity.

RESULTS AND DISCUSSION

As shown in Figure 1a,b, we successfully synthesized isolated nanosized iron precursors with a high degree of crystallinity, as evidenced by X-ray diffraction (XRD) and extended X-ray absorption fine structure spectroscopy (EXAFS). Under condition 1 (Figure 1c), all the samples obtained show the characteristic diffraction pattern of $\chi\text{-Fe}_5\text{C}_2$ (c.f. JCPDS no. 36-1248), indicating $\chi\text{-Fe}_5\text{C}_2$ is the dominant iron phase. But when under condition 2 (Figure 1d), the A-2 catalyst showed

catalyst, the Fe_3O_4 phase mainly lies in the shell while the iron carbides are predominant in the core. The boundary between the core and shell is more ambiguous for the A-1 than that of the C-1 catalyst. The different density of carbides compared with oxides (5.2 g/cm^3 for Fe_2O_3 and Fe_3O_4 vs 7.57 g/cm^3 for Fe_5C_2) might lead to the interlayer shrink for A-1 catalyst, while interlayer expansion for the C-1 catalyst. In other samples the TEM images under two conditions that show similar core-shell patterns can be found in Figure S1, and the specific compositions in each part of the particles are strongly dependent on the starting precursors. The peaks at ~ 286.0 and $\sim 296.0 \text{ eV}$ shown in the EELS profile (Figure 2) are ascribed to the C $\text{K}-\pi$ ($1s$ to π^* transition, which indicates the presence of sp^2 hybridized carbide species) and the C $\text{K}-\sigma$ ($1s$ to σ^* absorption peak, which stands for amorphous carbon species).^{19,20} For A-1 (Figure 2a,e), spot 1 shows the carbon $\text{K}-\sigma$ signal, the O K -edge signal as well as the Fe L-edge signal. Combining this with the lattice spacing from HRTEM, it implies that the iron oxide coexists with amorphous carbon species. For spots 2 and 3, C $\text{K}-\pi$, $\text{K}-\sigma$ signals, O K -edge, and Fe L-edge signals are observed, indicating the coexistence of iron carbide(s) and Fe_3O_4 phases in these regions. Note that the relative $\text{K}-\pi/\text{K}-\sigma$ intensity ratio ($I_{\text{K}-\pi}/I_{\text{K}-\sigma}$) increases along the spectrum scan line from spot 2 to spot 3, indicating the relative iron carbide composition increases along the core to the shell. For spot 4, only C $\text{K}-\sigma$ signals were observed, revealing the Fe particles are entangled by amorphous carbon deposits. For sample C-1 (Figure 2f,j), the main differences lie in the core region in Figure 2j from the iron carbide where the $I_{\text{K}-\pi}/I_{\text{K}-\sigma}$ decreases from the core to the shell (spot 5 to 6 and 7). The existence of amorphous FeC_x species can be confirmed by the fingerprint-like whorls across the whole region. X-ray photoelectron spectroscopy (XPS) (Figure 3a,b) shows the peaks related to Fe^{3+} ($\sim 711.3 \text{ eV}$), Fe^{2+} ($\sim 709.8 \text{ eV}$), and iron

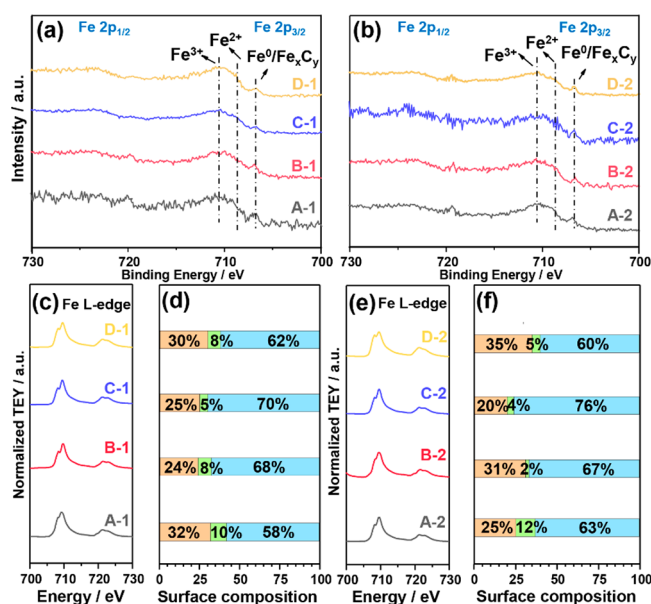


Figure 3. Surface characterization of catalysts under two conditions: (a,b) XPS spectrum of catalysts at two conditions, (c,e) Fe $\text{L}_{2,3}$ -edge X-ray absorption spectra, and (d,f) phase composition of the obtained samples determined by LCF method; in the bar graphs orange represents $\text{Fe}^{2+/3+}$, green represents Fe^{2+} , and blue represents Fe^0/FeC_x .

carbide species ($\sim 707.0 \text{ eV}$), meaning the surface of all the samples consist of iron oxide and iron carbide species, which are also confirmed by the Fe L-edge analysis. Together with previous XRD and XPS analysis, we propose the models (Figure 2d,i) for catalysts that are derived from oxidized and reduced iron precursors. Also, the bulk and surface species confirmed are of great significance to be employed as references for latter quantitative calculations.

Soft X-ray absorption spectroscopy was employed to characterize further the surface species of samples. The pre-edge of the O K -edge spectra at $\sim 532 \text{ eV}$ (Figure S2) indicates the presence of iron oxide species.^{25,26} The distinct splitting of the L_3 and L_2 edges by $\sim 1.5 \text{ eV}$ is due to the energy difference of the e_g level and t_{2g} level (Figure 3c,e).²⁷ We note that the intensity ratios of the t_{2g} feature to the e_g feature in the L_3 -edge for the samples are higher than that of the standard spectrum of the Fe_2O_3 L -edge but lower than that of the metallic iron reported in the literature, which indicates the presence of hybridized iron species.^{28–30} On the basis of the XPS and EELS results, quantitative analysis was performed using α -Fe, FeO , and Fe_3O_4 as references for Fe^0 , pure Fe^{2+} , and mixed ($1/2$) $\text{Fe}^{2+}/\text{Fe}^{3+}$ (Figure S3), with the characteristic main L_3 -edge contribution located at 709.3 eV for Fe^{3+} , 707.9 eV for Fe^{2+} , and 706.8 eV for $\text{Fe}^0/\text{Fe}_x\text{C}_y$.³¹ A similar linear combination fitting (LCF) method was also employed on Fe-based catalysts by de Smit et al.²⁵ The relative surface compositions are shown in Figure 3d,f. It is found that the average surface iron oxidation state (OS) of samples (Figure 4) stay in a narrow

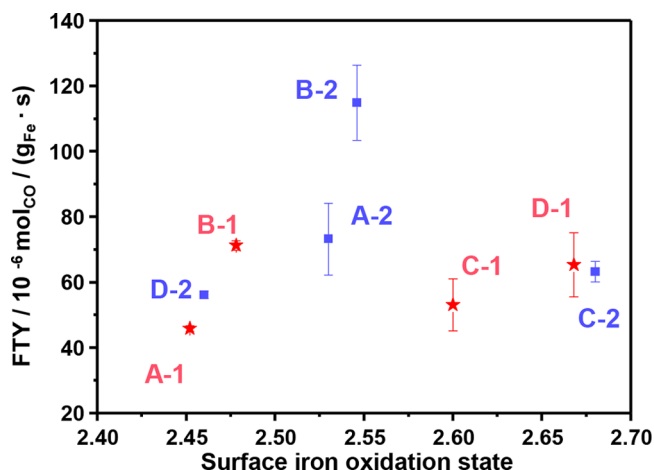


Figure 4. Catalysts FTY under different conditions as a function of surface iron OS.

range between $+2.40$ and $+2.70$. Conversely, the iron time yield (FTY, moles of CO converted to hydrocarbons per mol of Fe per second) varies greatly from 45.9 to $114.9 \times 10^{-6} \text{ mol CO g}_{\text{Fe}}^{-1} \text{ s}^{-1}$. Therefore, the surface OS cannot be used to interpret the variation of FTY reliably (Figure 4). During the FTS, the heterogeneous structure is formed under syngas atmosphere, which involves (i) CO dissociation and $\text{Fe}-\text{C}$ bond formation,^{32–34} (ii) carbon permeation and diffusion from surface to matrix,^{35,36} as well as (iii) structure reconstruction and phase transformation.²¹ Despite the controversy concerning the actual active sites, the correlation between the carbide content and activity had been widely reported, suggesting the important role of bulk properties in determining FTS performance.^{7,11,37} Raupp et al. proposed

that the bulk iron functioned as an obtainer for surface carbon, which influenced surface structure and, hence, the active site density.³⁸ In contrast, a competition model was proposed that surface “C-containing” species hydrogenation and iron carburization occur simultaneously as competitive reactions.³⁹ Bartholomew and co-workers reported that the initial catalytic activity is proportional to the amount of adsorbed atomic carbon on the surface, while the steady-state activity is related to carbon species at the surface and in the bulk of iron crystallites.⁴⁰ In recent work, Hoffmann and Wen et al. explored carbon permeation theoretically on different model surfaces and elucidated how it affects the Fe-based FTS, and carbon surface migration and subsurface permeation were found to take place simultaneously, but showing different preferences at various coverages on different crystal planes.³⁶ Although the link between surface and bulk phase of iron is still not clear, it is reasonable to postulate that both the surface and bulk properties should be taken into consideration for Fe-based FTS.

Recognizing this, we focus on analyzing the catalytic relevance of the bulk properties in these samples in the following sections. It is well acknowledged that there is a linear relationship between the average OS and the characteristic edge positions in the XANES spectra.^{41,42} Therefore, we employed several iron species as references (Figure S4) to correlate quantitatively the OS of Fe species and the characteristic edge positions. The calculated bulk OSs of two series of catalysts are summarized in Table S2. Moreover, based on previous surface and bulk characterizations analysis, Fe_3O_4 , FeC_x , and Fe_5C_2 are used as standards to calculate the relative compositions for samples from the XANES spectra (the details are provided in the Supporting Information). The phase map distribution results are displayed in Figure 5a–h. The quantitative analysis shows that most Fe species exist as carbides; only a small fraction of the Fe_3O_4 phase (<6.0%) was

detected. Moreover, the catalysts derived from various precursors are primarily identified through differences in their carbide species composition.

The catalytic testing results are summarized in Table 1. A general trend is observed: the higher the OS of iron species, the higher the FTY that can be achieved. Although the FTS has been recognized as a size-sensitive reaction, the particle sizes of iron species for all samples in our study are beyond the critical value (7 nm for an undoped catalyst).¹³ Therefore, the differences in activity and selectivity may originate from differences in their electronic properties. Notably, in Figure 6a, a linear relationship between FTY and bulk OS can be established, while the FTY versus surface OS does not reveal any clear trends (Figure 4). The OS of iron species can be considered as an indicator of the electron density of the active center, which would play an important role in the catalytic reaction. In this case, iron species with higher OS tend to be in the electronically deficient state,⁴³ possibly giving rise to an enhanced σ -coordination and a weakened π -back-donation for CO adsorption. As the contribution of π -back-donation to CO adsorption on Fe species is predominant which is confirmed via a red-shift in the Fourier Transform Infrared Spectroscopy,⁴⁴ it is reasonable that the higher OS results in a decreased strength of Fe–CO binding. On the basis of both experimental and DFT calculations,^{13,45} the weaker Fe–C bond could enhance FTS activity because of easier removal of C and O atoms from the surface. Moreover, numerous reports have revealed that $\chi\text{-Fe}_5\text{C}_2$, Fe_7C_3 , and Fe_2C species with higher iron OSs (+1.60, +1.71, and +2.0, respectively) are more active than carbides with lower iron OSs (i.e., Fe_4C of +1.0 and $\theta\text{-Fe}_3\text{C}$ of +1.33), which is also in line with our results.^{7,9,12,37} Therefore, although the attribution of active species is still disputable and multisite active centers may coexist, the OS of Fe species can be used as a universal descriptor to reflect the activity of CO hydrogenation.

On the other hand, as a least desired product in FTS, the formation of CH_4 via successive hydrogenation usually provides an indication of performance of the hydrogenation ability of a catalyst with respect to C–C coupling. We tried to correlate the FTY_{CH_4} to the chemical properties of Fe species, and fortunately it was found that the FTY_{CH_4} increases proportionally to the Fe–C coordination number (CN) (as displayed in Figure 6b). This result is in line with the theoretical study by Jiao's group, where a linear relationship between the effective barrier of CH_4 formation and d-band center of the surface was established.⁸ Generally, CH_4 formation can be understood as successive insertion of H atoms into the Fe–C bond. According to the principle of microscopic reversibility, the CH_4 formation thus could be easily understood by the activation of C–H, which is mainly guided by electron back-donation from Fe atoms into the antibonding σ_{CH^*} state of CH_x . In an iron carbide dominated catalyst system, as the Fe–C CN increases, the d-band center moves away from the Fermi energy, resulting in an enhanced C–H activation. Consequently, a lower effective barrier of CH_4 formation is then anticipated. Recently, the effective barrier differences between CH_4 formation and $\text{C}_1\text{--C}_1$ coupling (ΔE_{eff}) on different iron phases were calculated to describe CH_4 selectivity. The ΔE_{eff} follows the order: $\text{Fe} > \text{Fe}_3\text{C} > \text{Fe}_5\text{C}_2$,^{46,47} indicating the improved CH_4 selectivity with increasing Fe–C CN, which agrees with our work as well. Theoretical calculations revealed a linear relationship between

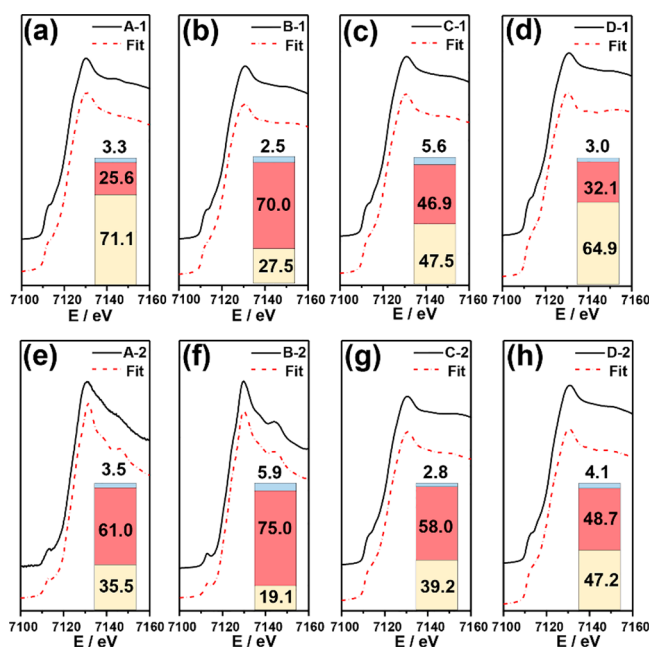


Figure 5. Fe K-edge XANES spectra and phase composition determined by LCF method of the samples obtained under condition 1 (a–d) and 2 (e–h); in the bar graphs yellow stands for Fe_3O_4 , red stands for FeC_x , and cyan stands for Fe_5C_2 .

Table 1. Catalytic Performance and Calculated Iron OS of Catalysts at Two Conditions

Sample	FTY (10^{-6} mol _{CO} g _{Fe} ⁻¹ s ⁻¹)	X_{CO} (%)	CO ₂ Sel. (%)	Hydrocarbon Sel. (%)						Average iron OS
				CH ₄	C ₂₋₄ ^a	C ₂₋₄ ^b	C ₅ ⁺	Oxy. ^a	O/P ^b	
A-1	45.9 ± 0.9	23.9	15.2	5.0	5.1	6.3	83.6	0.0	1.2	1.21
B-1	71.3 ± 1.4	66.5	32.5	8.4	19.1	20.5	51.5	0.5	1.1	1.62
C-1	53.1 ± 8.0	32.1	35.8	9.6	24.5	11.2	54.5	0.2	0.5	1.39
D-1	65.3 ± 9.8	8.1	5.3	3.4	4.3	2.1	90.2	0.0	0.5	1.25
A-2	73.2 ± 11.0	44.1	36.2	9.7	19.4	13.5	57.4	0.0	0.7	1.82
B-2	114.9 ± 11.5	24.3	20.7	5.5	18.8	14.4	60.9	0.4	0.8	2.00
C-2	63.3 ± 3.2	49.2	36.1	9.3	16.1	15.9	58.7	0.0	1.0	1.57
D-2	56.2 ± 1.1	48.6	36.6	8.9	17.2	19.4	54.5	0.0	1.1	1.44

^aOxygenates. ^bThe molar ratio of olefin to paraffin in the C₂₋₄ hydrocarbon.

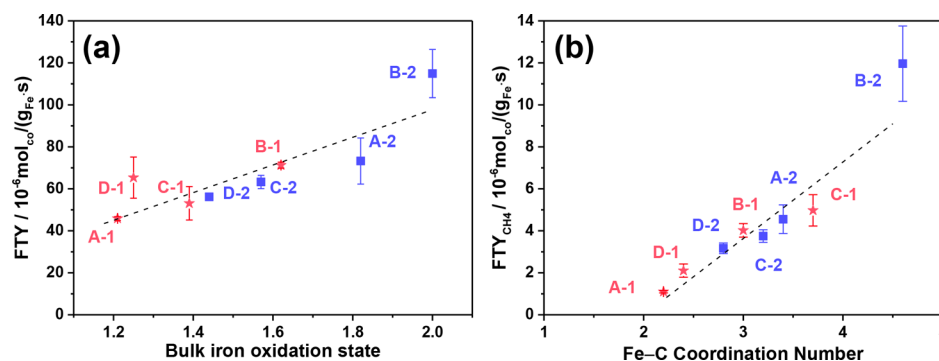


Figure 6. (a) FTY of catalysts as a function of bulk iron OS; (b) iron time yield of CH₄ (FTY_{CH₄}) as a function of Fe–C CN.

the effective barrier of CH₄ formation and the d-band center of the iron carbides species (i.e., Fe₂C (011), Fe₃C₂(010), Fe₄C (100), and Fe₃C (001), respectively),⁸ which are in accord with our results. de Jong et al. also found that increasing the electron density of iron species by introducing a promoter tended to produce less CH₄ and more lower olefins.¹⁴

Products distribution is highly dependent on the FTS CO conversion (X_{CO}), thus a fair comparison was made among catalysts under similar X_{CO} . With an X_{CO} of about 45%, A-2, C-2, and D-2 samples possess lower olefins selectivity of 13.5%, 15.9%, and 19.4%, respectively, and the relatively lower OS seems to play an important role in achieving a higher O/P molar ratio and lower olefins selectivity. This might be attributed to the weakened olefins adsorption and suppressed secondary hydrogenation on the higher electron-rich iron species, which have been confirmed by lower olefins (e.g., C₂H₄, C₃H₆) injection/pulse experiments on Fe catalysts with alkali promoters.^{48,49} These phenomena are also in good agreement with DFT calculations, which suggests that weak adsorption of olefins caused by electron donation inhibits their further hydrogenation, leading to high selectivity.⁴⁸ Above all, this phenomenon would be of great interest in our future studies.

CONCLUSION

In conclusion, eight samples produced from four pure Fe phases were obtained by changing the gaseous atmosphere and pressure. Multiple surface and bulk characterization techniques were employed to elucidate the impact of the chemical environment upon Fe species and on FTS performance. The core-shell structures are identified for all catalysts by TEM combined with EELS, but the compositions for different counterparts are strongly dependent on the properties of the iron precursors. On the basis of a precise quantification of the

different Fe related species, we found that catalytic activity and selectivity are generally governed by the bulk chemical environment of the Fe species rather than their surface properties. Despite different spatial contributions of the Fe species, the FTY is found to correlate linearly with the OS of the Fe species and the CH₄ yield is related to the CN of the Fe–C bond. Our work provides guidance for designing and modifying Fe-based FTS catalysts.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.9b01104.

Experimental section, particle size, XANES and EXAFS analysis, HRTEM (PDF)

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Notes

The authors declare no competing financial interest.

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Supporting Information

Elucidating surface and bulk phase transformation in Fischer–Tropsch synthesis catalysts and their influences on catalytic performance

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1. Experimental Procedures

1.1 Catalyst preparation

Nanoparticulate iron oxide (Fe_2O_3), iron carbides (Fe_5C_2 and Fe_3C) and metallic iron (Fe^0), denoted as A, B, C and D, respectively, were synthesized using the following procedures.

The Fe_2O_3 nanoparticle (NP) preparation procedure was adapted from Torres Galvis et.al.¹ Typically, 5.0 g of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (98% A.C.S. reagent, Sigma-Aldrich) were completely dissolved in 20.0 mL of distilled water. The solution was dried at 60°C for 6 h and then at 120°C overnight to isolate the precursor as a reddish brown solid. Finally, with a ramp rate of 5°C·min⁻¹, the precursor was calcined under air flow at 300°C for 5 h.

Fe^0 NPs preparation followed a method reported method in the literature by Zheng et al.² In a typical synthesis, 4.0 g $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ (Sigma-Aldrich, > 97%) was dissolved in 20.0 mL ethanol to give a solution. Subsequently, 5.0 g NaOH (Sigma-Aldrich, > 98%) and 10.0 mL $\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$ (Sigma-Aldrich, 78-82%) was added into the solution. The mixture was stirred vigorously to homogeneity and then transferred into an 80 mL stainless steel autoclave. The autoclave was sealed and kept at 80°C for 10 h and then passively cooled down to ambient temperature. Finally, the isolated product in ash black color was washed with distilled water and ethanol for thrice respectively, and then stored in ethanol for further characterization and catalytic testing.

Fe_3C NPs were synthesized though a biopolymer route according to the previously published literature.³ In a typical synthesis, 1.0 g of Iron-(II) acetate (Aldrich, 99.995%) was dissolved in 10 mL deionized water, then a few drops of H_2O_2 (Sigma-Aldrich, 50 wt. % in H_2O) were added, obtaining highly soluble iron(III) acetate solution (denoted as “solution A”). Then, 2 g of Gelatin (Type A from porcine skin, Sigma #G2500) was dissolved in deionized 100 mL water (100 mL) at 90°C, and a clear, colorless solution (denoted as “solution B”) was achieved. For the synthesis of iron carbide, 6 g of the prepared iron (III) acetate solution was added dropwise to 5 g of the prepared Gelatin solution under vigorous stirring, forming a clear, brown and viscous solution. Products obtained were cast to Petri dishes and dried at 50°C in air to give clear red/brown films. Agate mortar and pestle were then used to grind the dried films into an even state, and finally the products are calcined at 2°C /min to 650°C in lidded crucibles in a box furnace under N_2 and cooled

immediately, yielding ash black powder.

Fe₅C₂ NPs were prepared following a previously reported method.⁴ In a typical synthesis, 14.5 g of octadecylamine (Aldrich, GR, 90 %) and 0.113 g of CTAB (BDH LTD, 98%) was mixed sufficiently and degassed under Ar. With the mixture being heated to 120°C, 0.5 mL of Fe(CO)₅ (Aldrich, >99.99%) was injected under an Ar blanket. Then the mixture was further heated to 180°C with a ramp rate of 10°C/min and kept at this temperature for 10 min. After this, the solution was heated to 350°C at 10°C/min and kept at this temperature for 10 min before it was cooled down to ambient temperature. Finally, the product was washed with ethanol and hexane to remove the excess octadecylamine and the as-synthesized NPs were kept in hexane for further use.

Commercially available Fe₃O₄ (99.5%, 15-20 nm APS, nanostructured & amorphous materials inc., USA) was also purchased for XAS reference use.

1.2 Catalyst characterization methods

Prior to X-ray Diffraction characterizations, all samples were dried and grinded in an agate mortar and pestle in an argon atmosphere. A Bruker D8 Advance X-ray Diffractometer operating under Cu K α radiation (40kV, 40mA) equipped with a LynxEye detector was employed to obtain XRD patterns. The samples were scanned over the 2 θ range 10° to 90° with a step size of 0.02° and a count time of 3.2 seconds per step.

High resolution transmission electron microscope (HRTEM) was performed using a Tecnai G2 F20 system equipped with field-emission gun at 200 kV. Electron energy loss spectroscopy (EELS) analyses were carried out in the scanning transmission electron microscopy (STEM) mode with a high-angle annular dark field (HAADF) detector, and images were collected using a Gatan Ultrascan 2k*2k CCD camera with a beam resolution of 0.2 nm. All data collection was conducted swiftly with very short acquisition times to prevent changes caused by electron-beam irradiation. Data processing and analyses were conducted by using Gatan Digital Micrograph software. Samples were grinded to powder and then dispersed homogeneously in ethanol by ultrasonication, and finally 2 drops of suspension were dropped onto a carbon-coated copper grid. Note that the obtained EELS spectra are 2D projection of the 3D objects, which would contain both surface and bulk information.⁵ The particle size histograms were obtained from TEM analysis using at least 300

nanoparticles.

X-ray photoelectron spectroscopy (XPS) and Fe near-edge X-ray absorption spectra (NEXAFS, Fe L-edge analysis) measurements were performed on the Soft-X-ray beamline at the Australian Synchrotron (AS). A Kratos Axis UltraDLD equipped with a SPECS Phoibos 150 hemispherical analyzer was used. Samples were firstly bushed onto Kapton tape that stuck to gold plate in the glove-box, and then transported to the load-lock and then to the sample parking chamber through the fast sample entry. A clean and ultra-high vacuum (UHV, base $\sim 6 \times 10^{-9}$ mbar, working $\sim 5 \times 10^{-6}$ mbar) environment was maintained for the Soft X-ray beamline and end-station operation. Spectra were excited using a monochromatized Al K α X-rays (1486.7 eV) with the X-ray source operating at 150 W. The electron analyzer was set with a resolution of 0.1 eV with a pass energy of 10 eV and a dwell time of 0.5 s for core level scans, and the adventitious metallic Au 4f 7/2 at 84.0 eV was taken as an inner calibration standard. Data collection and processing were conducted by using the SpecsLab software. Fe O K-edge and L-edge data were collected using the total electron yield (TEY) mode by measuring the drain current of the sample as a function of photon energy. The TEY absorption signal (I_1) was normalized by the photocurrent induced on a gold-mesh reference monitor (I_0). The QANT program operated in Igor Pro software has been used to view the corrected data channel,⁶ as well as to normalize the TEY absorption signal (I_1) by the photocurrent induced on a gold-mesh reference monitor (I_0). The second feature in the L₃ edge was calibrated to 709.4 eV. The final spectra were presented by the average of three scans to exclude the possible spurious information, and finally the spectra were processed by the CTM4XAS program.⁷

Samples was ground and diluted with proper amount of boron nitride to achieve a uniform state, and then the mixture was pressed to a 13 mm pellet by a hand-held press, and the pellet was finally mounted into the PMMA sample holders between two layers of Kapton tape. To minimize the possible re-oxidation of catalysts for structural characterization, all procedures mentioned above were manipulated in the glovebox.

The Fe K-edge XAFS were performed on the 20B beamline at AS and collected in the transmission mode in the cryostat at 10K. The beamline is equipped with Si (111) and Si (311) monochromators and operates in the range of energies between 5 and 31 keV. In our experiment, the X-rays were passed through a Si (111) crystal monochromator and focused onto the sample. The Athena and Artemis program were used to extract and process the data.⁸⁻⁹ The X-ray absorption

edge energy was calibrated by using the absorption edge of metallic iron foil. For the Fe K-edge XANES spectra, data were obtained in the range from 6.912 to 7.162 keV with a spot size of approximately 0.25 * 0.25 mm. After background subtraction and normalization procedures, experimental absorption coefficients as a function of energies $\mu(E)$, denoted as “normalized absorption”, were obtained. Three consecutive scans were run for each sample to improve the spectrum quality. Besides, principle components analysis (PCA) was employed to determine the number of Fe species that contribute to the perfect fit of spectral features. Linear combination fitting (LCF) method, as well as the collected spectra of references, were used to estimate the percentage of each composition. For the EXAFS part, the k range measured was 0 – 15.0 Å. The amplitude reduction factor, S_0^2 , was calculated by fitting the standard Fe foil, which is fixed in all cases. By fitting the EXAFS data in k and R space, local structural parameters including coordination number (CN), bond length (R), Debye–Waller factor around the absorbing atoms and inner potential correction (ΔE_0) were acquired. One to note is, another two randomly selected spots were scanned for each sample to avoid noise disturbance. The data overlaid perfectly with each other for all samples, indicating the evenness of samples.

1.3 Catalytic reaction

Catalytic reaction was evaluated under similar process conditions (feed composition and pressure). To avoid products condensation, a N₂ flow was used to blow downstream each of the reactor flow. Feed gas was split equally and after flowing through the catalyst bed in the reactor, and an Agilent 7890A online gas chromatography equipped with two parallel GC channels was used for the effluent online analysis: Front channel connected with an Agilent PoraBOND Q packed column (CP7352) with a TCD detector for Ar, CH₄ and CO analysis, and Auxiliary channel connected with an Agilent MolSieve 5A packed column (G3591-80046) with a TCD detector for CO₂ and C₁–C₄ components separation.

FTS experiments were carried out by a Flowrence high-throughput unit (Avantium®), equipped with 48 parallel fixed-bed microreactors set in 3 separate temperature blocks. This equipment has been previously described by Lippi et al.¹⁰ The catalytic reaction was performed under the conditions of 20 bar, 340°C, WHSV=30,000 mL/(g·h) consisting of CO 45 vol. %, H₂ 45

vol. % and Ar 10 vol. % (Argon incorporated as an internal standard). After reaction, samples were cooled to room temperature and collected for characterizations under the inert atmosphere, via a custom-made container connected to N₂ cylinder to transport the samples from the Flowrence rig to the glovebox, with the aim of protecting the air sensitive surface of samples. The catalytic performance was averaged by 4 points. CO conversion (X_{CO}, %) and products selectivity (%), calculated on carbon basis with CO₂ not included) were employed to obtain the catalytic performance as following expressions:

$$X_{CO} = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} * 100\%$$

$$C_nH_m \text{ selectivity} = \frac{n * C_nH_m_{outlet}}{CO_{inlet} - CO_{outlet} - CO_{2outlet}} * 100\%$$

$$CO_2 \text{ selectivity} = \frac{CO_{2outlet}}{CO_{inlet} - CO_{outlet}} * 100\%$$

2. Results and Discussion

Table S1. Particle sizes of catalysts at different conditions

Sample	A / nm	B / nm	C / nm	D / nm
Precursor ^[a]	11.8	16.5	25.1	54.1
Condition 1 ^[a]	23.6	18.6	26.7	26.1
Condition 1 ^[b]	22.2±3.5	19.0±8.5	25.0±7.6	24.2±5.5
Condition 2 ^[a]	17.5	16.0	22.9	18.6
Condition 2 ^[b]	18.5±4.8	17.5±5.6	23.2±6.8	18.8±4.2

[a] Calculated by Scherrer equation; [b] Estimated from TEM images.

Table S2. Energy shifts of various spectral features in the Fe K-edge XANES spectra of iron catalysts (± 0.1 eV) and calculated oxidation states results

Sample	Averaged		Energy	Energy	Energy	Oxidation	Oxidation	Oxidation
	iron	Energy shift	shift	shift	shift	state	state	state
	oxidation	(Threshold)	(Pre-	(main	(White	calculated	calculated	calculated
	states ^a		edge)	edge)	line)	by	by main	by white
						threshold	edge	line
A-1	1.21	0.42	N.A.	10.5	20.1	1.21	1.19	1.24
B-1	1.62	0.69	N.A.	11.2	20.5	1.61	1.59	1.67
C-1	1.39	0.55	N.A.	10.9	20.2	1.40	1.42	1.35
D-1	1.25	0.45	N.A.	10.6	20.1	1.25	1.25	1.24
A-2	1.82	0.81	2.9	11.7	20.6	1.80	1.88	1.78
B-2	2.00	0.95	2.4	11.9	20.8	2.01	1.99	2.00
C-2	1.57	0.68	N.A.	11.2	20.4	1.60	1.53	1.57
D-2	1.44	0.58	N.A.	11.1	20.2	1.45	1.48	1.40

a: Calculated by averaging column 7, 8 and 9;

*N.A.: Not available.

Table S3. EXAFS model fitting

Sample	Shell	CN ^[a]	R / Å ^[b]	σ^2 / Å ² ^[c]	ΔE_0 (eV) ^[d]	R-factor ^[e]
A-1	Fe–C	2.2	2.035	0.0127	-4.569	0.015
	Fe–Fe	3.4	2.531	0.0227		
	Fe–Fe	2.0	2.658	0.0042		
B-1	Fe–C	3.0	2.056	0.0150	-3.368	0.015
	Fe–Fe	3.4	2.531	0.0124		
	Fe–Fe	1.7	2.656	0.0207		
C-1	Fe–C	3.7	2.069	0.0163	-2.891	0.014
	Fe–Fe	4.4	2.528	0.0153		
	Fe–Fe	5.8	2.659	0.0124		
D-1	Fe–C	2.4	2.037	0.0162	-3.121	0.022
	Fe–Fe	4.0	2.530	0.0150		
	Fe–Fe	5.3	2.661	0.0155		
A-2	Fe–O	2.8	1.956	0.0182	-7.246	0.025
	Fe–C	3.4	2.058	0.0125		
	Fe–Fe	3.8	2.731	0.0121		
B-2	Fe–C	4.6	2.035	0.0196	-7.085	0.030
	Fe–Fe	4.7	2.543	0.0228		
	Fe–Fe	5.5	2.665	0.0251		
C-2	Fe–C	3.2	2.060	0.0113	-2.169	0.028
	Fe–Fe	4.7	2.523	0.0142		
	Fe–Fe	6.0	2.652	0.0140		
D-2	Fe–C	2.8	2.048	0.0179	-3.581	0.022
	Fe–Fe	4.6	2.541	0.0176		
	Fe–Fe	6.6	2.660	0.0173		

[a] CN, coordination number; [b] R, distance between absorber and backscattered atoms; [c] $\Delta\sigma^2$, Debye–Waller factor; [d] ΔE_0 , inner potential correction; [e] R-factor, closeness of the fit.

The Debye-Waller factor (see [Table S3](#)) measures for the thermal, static and dynamic disorder in the local scale. For B-1 catalyst, the comparatively larger Debye-Waller factor compared with that of B-2 catalyst suggest the Fe_5C_2 catalyst is more susceptible to be oxidized into disordered structure under FTS conditions in the local scale, which agrees with the conclusion drawn by de Smit et.al.¹¹ Furthermore, for Fe_2O_3 samples, the obvious EXAFS spectra differences between two conditions that changed from iron carbide-like to iron oxide-like pattern, i.e. the disappearance of Fe-Fe bond at ~ 2.5 Å corresponding to iron carbide in A-1 catalyst, and the appearance of Fe-O bond at ~ 2.0 and ascribed to the oxide feature in A-2 catalyst, evidently indicate that the long-range order and local environment underwent phase transformation, probably result from the amorphous FeC_x phase oxidation.

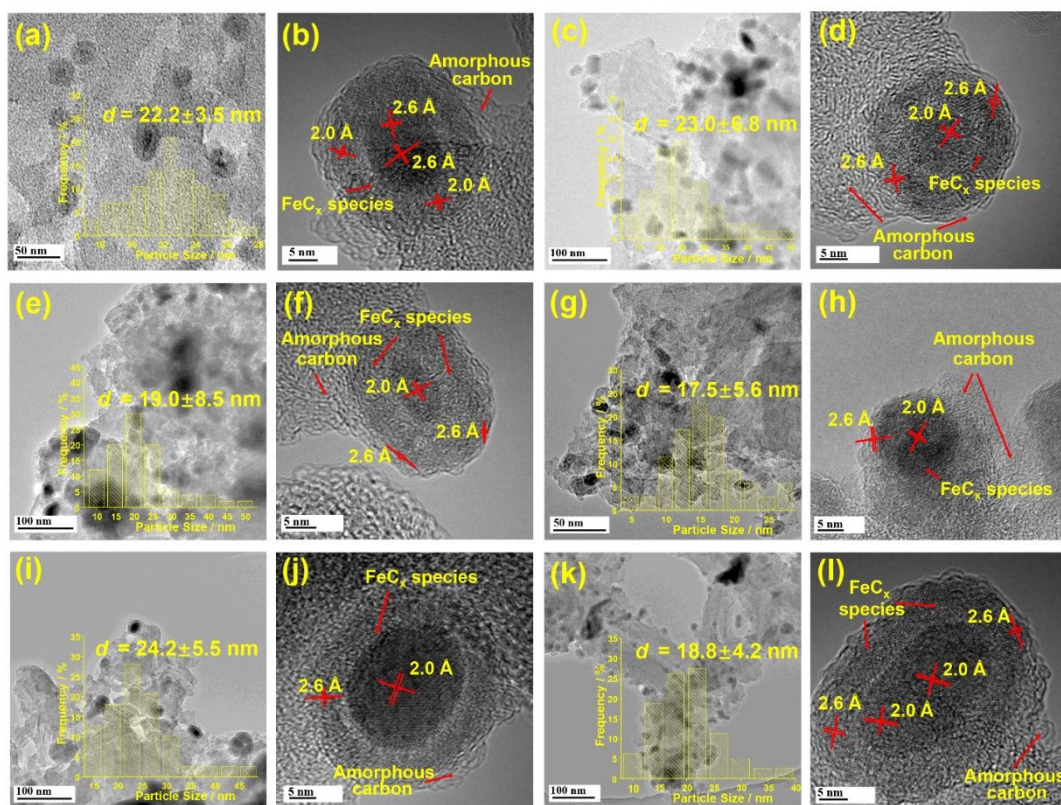


Figure S1. High-resolution TEM images of core-shell structured catalysts: a-b) A-2; c-d) C-2; e-f) B-1; g-h) B-2; i-j) D-1; k-l) D-2

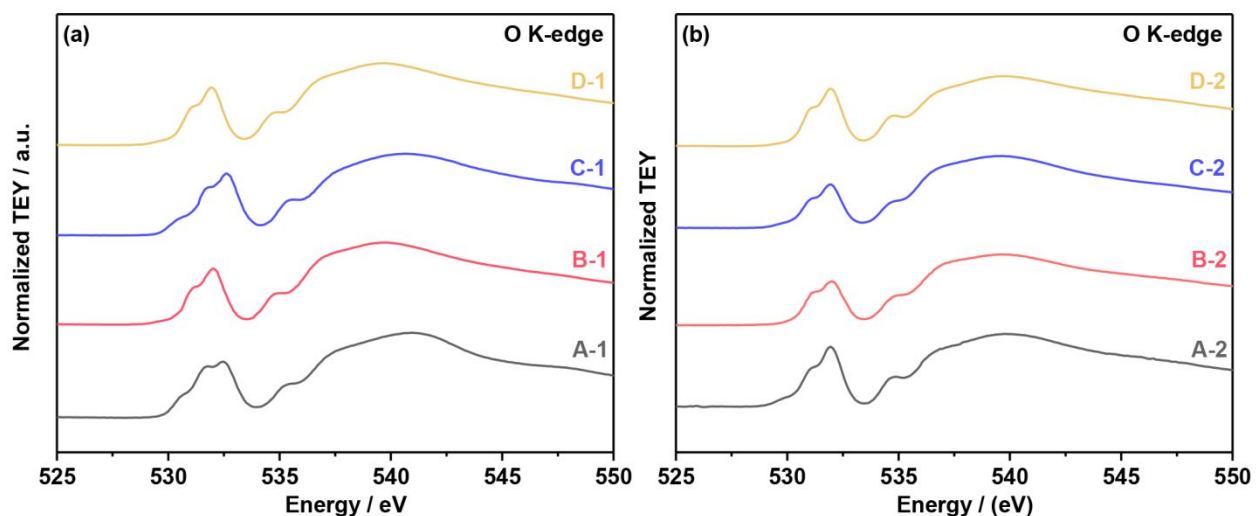


Figure S2. Soft X-ray absorption spectra of O K-edge spectra of samples at two conditions

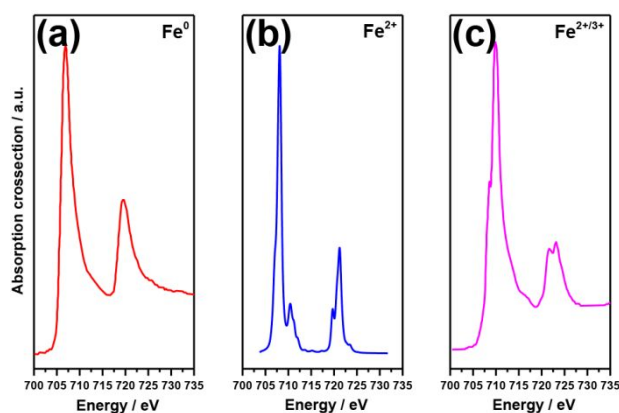


Figure S3. Theoretically simulated Fe L₃-edge absorption spectrum for a) Fe⁰, b) pure Fe²⁺ and c) mixed (1/2) Fe²⁺/Fe³⁺.

The O K-edge and Fe L_{2,3}-edge spectra were collected, and charge-transfer multiplet (CTM) simulations with crystal-field multiplet theory incorporated into an atomic multiplet calculation were carried out to gain more informative details about surface compositions.¹²⁻¹⁷

One to note is, α -Fe and Fe₃O₄ were adopted from the previous collected data at AS,¹⁸ and octahedral (O_h) Fe²⁺ ion was simulated with a configuration of O_h geometry, with the ionic value

($10Dq(\text{ion})$) of -1.0 eV, charge transfer energy (Δ) of 4, $t_{eg}=1$ and $t_{2g}=2$,¹⁹ which is found to match reported FeO data well.^{13, 20}

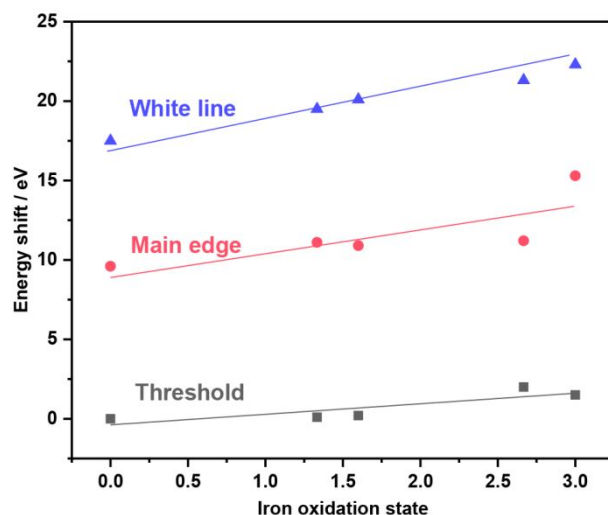


Figure S4. Linear relationships between the averaged iron oxidation state and energy shifts of various spectral features (with iron oxidation state of 0.0 for metallic Fe, +1.33 for Fe_3C , +1.60 for Fe_5C_2 , +2.67 for Fe_3O_4 and +3.0 for Fe_2O_3)

The positions of the edge features are sensitive to the metal oxidation states, which can be understood by the Coulomb effect. Generally, the zero of energy is confirmed by locating the first inflection point of the iron metal in the derivative spectrum at 7110.3 eV, which marks the threshold of photo-ejection of the 1s electron in metallic iron.

Then, absorption features of samples under two conditions, including the energy of a) the absorption threshold (defined by position of the first peak in the derivative curve); b) pre-edge peak; c) the absorption edge (defined by position of the second peak in the derivative spectrum), and d) the 1s to 4p transition above the absorption edge as the predominant peak (white line), had been collected to facilitate further calculation. Finally, the energy shifts of absorption features with respect to zero of energy, are enumerated in [Table S3](#) (see column 3, 4, 5, and 6), respectively.

According to Kunzl's law, the energy shifts of the references vary linearly with the oxidation state of the absorbing iron atom (see [Figure S4](#), three model linear function determined as best fits to the data from five references). Then, by regressing the energy shifts in column 3, 5, and 6 of

Table S3, respectively, to the linear functions shown in Figure S4, iron oxidation states of the samples are calculated (see column 2 in Table S3). Clearly, no remarkable differences were observed for the oxidation states calculated from three linear function (see column 7, 8 and 9 in Table S3), which evidence the feasibility of this calculation method.

PCA and LCF methods had been used to examine the relative abundance of various iron species in the near-edge spectral region. In our work, the long-range order (XRD patterns in the main text, Figure 1(c-d)) structure coincide well with that confirmed by local-range order (EXAFS spectra shown in Figure S6) analysis.

Similarities in Figure 1(c-d) and Figure S6 between various samples indicate similar references can be chosen to fit the A-1, B-1, B-2, C-1, D-1 and D-2 XANES pre-edge features. We identified by PCA that three major iron components, i.e. Fe_3O_4 , FeC_x , (resembled by Fe_3C as more reduced iron species, since the EXAFS analysis showed a somewhat lower contribution of C neighbors compared to Fe_5C_2 reference, and the LCF of samples only yielded perfect fits after including the ‘ Fe_3C ’ reference contribution.) and Fe_5C_2 , involved in the reaction. Metallic Fe and Fe_2O_3 were specifically rejected by the PCA, and attempts to include Fe^0 and Fe_2O_3 in the LCF did not improve the spectral fits. Because of the similarity between XANES spectra of iron carbides species, the uncertainty of species is inevitable. Herein, Fe_3O_4 , FeC_x , and Fe_5C_2 are reasonably used as references for our LCF.

For A-2 sample, EXAFS patterns clearly show characteristic Fe–C, Fe–O and Fe–O–Fe bonds, respectively, which are quite likewise that of $\alpha\text{-Fe}_2\text{O}_3$ catalysts under high carbon chemical potential (μ_{C}) FTS conducted by Weckhuysen’s group.¹¹ Therefore, FeC_x (represented by ‘ $\alpha\text{-Fe}$ ’), as well as Fe_5C_2 , Fe_3O_4 were employed in LCF to obtain a high-quality fitting for these samples.

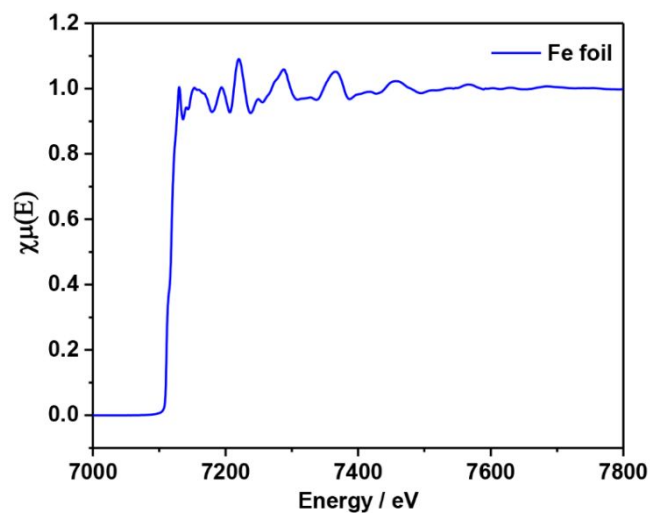


Figure S5. Normalized absorption coefficient as a function of X-ray energy of a Fe foil K-edge.

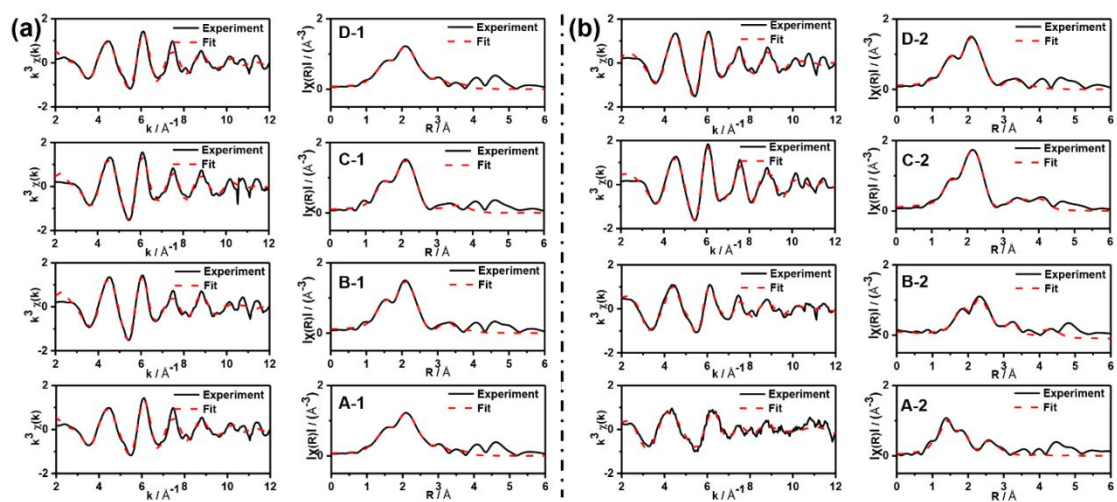


Figure S6. Non-phase corrected experimental (black, solid) and simulated (red, dashed) EXAFS spectra at two conditions plotted in k (left) space and R (right) space a) condition 1; b) condition 2.

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